Year 11 Set 2 · Higher · grade 5–6 · Lesson 4

Trigonometry (SOH CAH TOA & exact values)

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Today I will learn to...

Goals

1. Find missing sides with SOH CAH TOA

2. Find missing angles and use exact trig values

Do now: retrieval practice

4 mins

1. Hypotenuse of a triangle with sides 5 and 12

   13

2. Solve x/4 = 7

   x = 28

3. Round 6.4279 to 3 sf

   6.43

4. Solve 8 = 20/x

   x = 2.5

Objective 1

Find missing sides with SOH CAH TOA

Key idea

Learn

!

Label O, A, H from the angle. Choose sin, cos or tan from the two sides involved. If the unknown is on the bottom, divide.

Objective 1: Find missing sides with SOH CAH TOA

I do: example 1

I do

Angle 35°, hypotenuse 10 cm. Find the opposite.

1. Label: H = 10, want O
2. O and H → SOH: $\sin\theta = O \div H$
3. $O = 10 \times \sin 35°$
4. O = 5.74 cm (2 dp)

I do: example 2

I do

Angle 50°, adjacent 8 cm. Find the opposite.

1. Label: A = 8, want O
2. O and A → TOA: $\tan\theta = O \div A$
3. $O = 8 \times \tan 50°$
4. O = 9.53 cm (2 dp)

I do: example 3

I do

Angle 28°, opposite 6 cm. Find the hypotenuse.

1. Label: O = 6, want H
2. O and H → SOH: $\sin\theta = O \div H$
3. $H = 6 \div \sin 28°$
4. H = 12.78 cm (2 dp)

Mini whiteboards

We do

1. Hyp 12, angle 40°. Adjacent (2 dp)?

Answer: 9.19

2. Adj 5, angle 60°. Hypotenuse?

Answer: 10

3. Hyp 8, angle 30°. Opposite?

Answer: 4

4. Adj 10, angle 35°. Opposite (1 dp)?

Answer: 7.0

You do: choose your challenge

You do

Bronze

Hypotenuse 15 cm, angle 30°. Find the opposite.

Silver

Adjacent 9 cm, angle 55°. Find the opposite (1 dp).

Gold

A 40 m kite string makes 62° with the ground. How high is the kite (1 dp)?

Platinum

A 6 m ladder makes 72° with the ground. How far is its foot from the wall, and how high does it reach (1 dp)?

You do: answers

Check

Bronze

7.5 cm

Silver

12.9 cm

Gold

40 sin 62° = 35.3 m

Platinum

Foot 1.9 m, height 5.7 m

If you finish early

Extra

Bronze

Hypotenuse 20 cm, angle 60°. Find the adjacent side.

10 cm

Silver

Opposite 8 cm, angle 25°. Find the hypotenuse (1 dp).

8 ÷ sin 25° = 18.9 cm

Gold

A 5 m ramp makes 12° with the ground. How high does it rise (2 dp)?

5 sin 12° = 1.04 m

Platinum

From 30 m away, the angle of elevation to the top of a tree is 28°. Eye level is 1.5 m above the ground. How tall is the tree (1 dp)?

30 tan 28° + 1.5 = 17.5 m

Exam question 1

2 marks

A right-angled triangle has an angle of 42° and adjacent side 7.5 cm. Find the opposite side to 1 dp.

Space for working

Mark scheme: 7.5 × tan 42° = 6.8 cm

Exam question 2

2 marks

A ramp 4 m long makes an angle of 12° with the ground. How high is it? (2 dp)

Space for working

Mark scheme: $4 \times \sin 12° = 0.83$ m

Objective 1 achieved

Find missing sides with SOH CAH TOA



Objective 2

Find missing angles and use exact trig values

✓ Achieved: Find missing sides with SOH CAH TOA

Key idea

Learn

!

Use the inverse button: $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$. Learn exact values for 0°, 30°, 45°, 60° and 90° for the non-calculator paper.

Objective 2: Find missing angles and use exact trig values

I do: example 1

I do

Opposite 5, hypotenuse 9. Find the angle.

1. Label: O = 5, H = 9
2. O and H → SOH
3. $\sin\theta = 5/9$
4. $\theta = \sin^{-1}(5/9) = 33.7°$

I do: example 2

I do

Adjacent 4, opposite 7. Find the angle.

1. Label: O = 7, A = 4
2. O and A → TOA
3. $\tan\theta = 7/4$
4. $\theta = \tan^{-1}(7/4) = 60.3°$

I do: example 3

I do

Exact values to learn

1. $\sin 30° = \cos 60° = \frac{1}{2}$
2. $\sin 60° = \cos 30° = \sqrt{3}/2$
3. $\tan 45° = 1$, $\tan 60° = \sqrt{3}$

Mini whiteboards

We do

1. Adj 6, hyp 10. Angle (1 dp)?

Answer: 53.1°

2. tan 45° = ?

Answer: 1

3. Opp 5, hyp 10. Angle?

Answer: 30°

4. sin 60° exactly?

Answer: √3/2

You do: choose your challenge

You do

Bronze

Opposite 4, adjacent 4. Find the angle.

Silver

Opposite 7, hypotenuse 12. Find the angle (1 dp).

Gold

A ramp rises 1.2 m over a horizontal distance of 9 m. Find its angle (1 dp).

Platinum

Without a calculator, find $\cos 30° \times \tan 60°$.

You do: answers

Check

Bronze

$45°$

Silver

$35.7°$

Gold

$\tan^{-1}(1.2/9) = 7.6°$

Platinum

$\sqrt{3}/2 \times \sqrt{3} = 3/2$

If you finish early

Extra

Bronze

Opposite 9, adjacent 9. Find the angle.

45°

Silver

Adjacent 7, hypotenuse 15. Find the angle (1 dp).

$\cos^{-1}(7/15) = 62.2°$

Gold

A 4 m ladder reaches 3.6 m up a wall. Find its angle with the ground (1 dp).

$\sin^{-1}(0.9) = 64.2°$

Platinum

Without a calculator, find sin 30° + cos 60° + tan 45°.

½ + ½ + 1 = 2

Exam question 1

2 marks

A right-angled triangle has opposite 8 cm and adjacent 11 cm. Find the angle to 1 dp.

Space for working

Mark scheme: $\tan^{-1}(8/11) = 36.0°$

Exam question 2

2 marks

Without a calculator, find the value of sin 30° + cos 60°.

Space for working

Mark scheme: ½ + ½ = 1

Objective 2 achieved

Find missing angles and use exact trig values



Real life

Apply

You stand 50 m from the base of a tower. The angle of elevation to the top is 32°. How tall is the tower? (1 dp)

50 × tan 32° = 31.2 m

Extension: stretch yourself

Stretch

1. **Hypotenuse 10, angle 60°. Find the exact opposite side.**

$10 \times \sqrt{3}/2 = 5\sqrt{3}$

2. **An isosceles triangle has sides 8, 8 and 10 cm. Find a base angle (1 dp).**

$\cos^{-1}(5/8) = 51.3°$

Exit ticket: exam questions

Exit

1. Hypotenuse 20 cm, angle 38°. Find the opposite side to 1 dp. **(2 marks)**

2. Adjacent 10 cm, hypotenuse 14 cm. Find the angle to 1 dp. **(2 marks)**

3. Without a calculator, find tan 45° + sin 30°. **(2 marks)**

Exit ticket: mark scheme

Mark it

Q1 20 sin 38° = 12.3 cm [2]

Q2 $\cos^{-1}(10/14) = 44.4°$ [2]

Q3 1 + ½ = 1.5 [2]

Total: ____ / 6

Today I achieved

- ✔ **Find missing sides with SOH CAH TOA**
- ✔ **Find missing angles and use exact trig values**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Hypotenuse 18 cm, angle 25°. Find the adjacent side to 1 dp. [2]

2. Opposite 11 cm, angle 40°. Find the adjacent side to 1 dp. [2]

3. Opposite 5 cm, adjacent 12 cm. Find the angle to 1 dp. [2]

4. A plane flies 3 km along a path that climbs at 15°. How much height does it gain, in metres? [3]

5. Without a calculator, find sin 60° × cos 30°. [2]

Extra practice: Maths Genie → Edexcel question bank → SOHCAHTOA · Exact trig values